

Assessing the Impact of the 21st Century Assistive Technology Act on Technology Access for Disabled Americans

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ABSTRACT

Technology plays a crucial role in modern society, yet is often inaccessible to many disabled individuals. This study evaluates the impact of the 21st Century Assistive Technology Act, a policy designed to improve access to assistive technology. Because the policy is relatively new, little research has been done evaluating its effectiveness. To address this research gap, we analyzed changes in post-policy assistive technology access, examining funding, assistive device loans, and device and internet usage among disabled individuals. We used multiple regression analysis to establish a correlation between these variables, a two-way fixed effects model to control for time-invariant and between-state factors in causal analysis, and regression discontinuity to further examine the relationship between state funding and assistive technology loans. Our analysis shows that the reauthorization of the Assistive Technology Act increased state grants, which is causally linked with an increase in device loans. While we found that the policy did not significantly impact digital access for disabled Americans, we hypothesize that its continued implementation will improve these metrics in the near future.

Keywords: digital access, disabilities, device loans, internet access, multiple regression, fixed effects, regression discontinuity

I. INTRODUCTION

In 2022, over 2.5 billion people worldwide needed access to at least one assistive technology product (WHO, 2022). In the U.S., around 20% (more than 54 million) of Americans have a physical or mental disability and need access to assistive technology (Su & Mejía, 2024). Assistive technology is an umbrella term that includes any equipment or system designed to improve daily functionality for physically and mentally disabled individuals. This includes vision-assistance devices, hearing aids, and mobility devices such as wheelchairs (WHO, 2022). The first Assistive Technology Act (Foundational Tech Act) was enacted in 1988 under Reagan's administration. The act was reauthorized five times, most recently in 2022 (ATAP, 2022). This reauthorization was extremely important, as it increased grant allocation for state programs, supported data collection efforts, and gave states greater flexibility in using funds. The policy aimed to increase access to assistive technology devices and services for disabled individuals, upgraded to newer technology, and implemented a coordinated nationwide network of state programs. The program also evaluated individuals to provide personalized assistive technology with an emphasis on serving older adults (US Senate, 2022).

Assistive technology is tremendously important for the well-being of individuals with disabilities. A 2005 study claimed further research was needed in order to show the importance of increasing assistive technology access. The authors highlight its impact on disabled individuals' daily activities, social engagement, and employment. The paper explains that although assistive technology facilitates independence and decreases stress, lack of affordability and accessibility remain major environmental barriers to usage (Scherer et al., 2005). Additionally, a 2012 study explored recent trends in assistive technology for mobility, focusing on how these tools can enhance mobility and provide widespread benefits. However, this paper overlooks the key role that policy plays in improving technology accessibility. As the authors note, assistive technology cannot be transformative if it is inaccessible

(Cowan et al., 2012). A 2016 study compared disabled individuals who used assistive technology to those who did not and found statistically significant differences in mobility. Successfully performing actions such as walking three blocks and standing for 20 minutes was strongly associated with assistive technology use (Clay & Alston, 2016). Furthermore, assistive technology access boosts individual and societal well-being by increasing economic growth, reducing hospital visits, and facilitating social participation (Smith et al., 2022).

As digital technology and internet access continue to advance, the digital divide has become an even more prevalent issue. First-level digital divide refers to internet access, second-level refers to the skill of internet usage, and third-level refers to the benefits gained from using digital technologies (Ragnedda & Rui, 2017). Especially for disabled individuals, the first-level digital divide has posed significant barriers to internet access and technology usage. Without breaking the first-level barriers of internet access, the second and third-level barriers cannot be addressed, leading to a persisting digital divide.

A 2006 study asserts that physical or mental limitations may prevent disabled individuals from using digital technology or the internet to their full potential. Internet usage and the extent of usage for disabled people were compared to that of the rest of the population. There are several barriers to information and communication technology usage for disabled people, such as physical inability to use technology for those with sensory disabilities and high expenses (Dobransky & Hargittai, 2006). Since 2006, the Assistive Technology Act has undergone reauthorizations and improvements, so it is still unclear whether barriers to digital technology access remain for disabled individuals. More inclusive social policies are crucial to increase assistive technology access (Bradley & Poppen, 2003).

Most existing studies on technology access for disabled individuals were conducted before 2022, and little research has explored the impact of the 2022 reauthorization of the Assistive Technology Act. Because updates to the policy are relatively recent, its effects have not been widely studied, and further investigation must be done to determine its significance post-policy. In addition, prior research mainly examined qualitative survey-based data, while our research uses panel data to support quantitative analysis. The motive of our research is to assess the efficacy of the reauthorization in improving access to assistive technology, digital technology, and the internet for disabled Americans. Our research fills a gap in the field because close to no quantitative evaluation has been conducted regarding the 21st Century Assistive Technology Act. It is important to note that our research has several limitations, which are discussed in detail in the Discussion/Conclusion section.

Our research examines how the 21st Century Assistive Technology Act has affected access to assistive and digital technologies for disabled Americans through the lens of technological and disability welfare policy. The key input variable is a binary indicator for policy implementation: 0 for observations before 2022 and 1 for those after 2022. We assess four state-level outcome variables: (1) total number of assistive technology device loans, (2) state grant allocations, (3) access to digital devices (desktops, laptops, tablets), and (4) internet access. Device loans and grant allocations were selected as direct indicators of policy implementation and reach within assistive technology programs. Additionally, digital device and internet access were included as broader measures of progress in bridging the digital divide for Americans with disabilities. We ran three models – multiple regression, fixed effects, and regression discontinuity – to evaluate the correlational and causal relationships between those variables.

We aim to address this central research question: Has the 21st Century Assistive Technology Act improved access to assistive and digital technology for Americans with disabilities?

Our research findings will contribute to existing research surrounding assistive technology and cultivate a deeper understanding of how recent policies have impacted disabled individuals' access to technology. By evaluating the quantitative outcomes of the 21st Century Assistive Technology Act on technology access, we aim to inform future initiatives in closing the technology access gap for individuals with disabilities.

II. METHODOLOGY

A. Data collection strategy

We collected data from multiple sources to evaluate assistive technology access. Device loan data by state was extracted from the Center for Assistive Technology Act Data Assistance (CATADA). We used this data as a metric for assistive device access for disabled individuals. While the data does not include the actual proportion of disabled individuals with access to assistive technology, it is still a strong indicator of overall assistive device access as a result of state programs. Federal grant allocation data was sourced from the Administration of Community Living (ACL). We collected this data in order to quantify the state-level financing provided in the act itself. The website provides data on federal grant allocations for state-level assistive technology programs from 2016 to 2024. To evaluate the policy's impact on the digital divide, we used panel data from the Computer and Internet Use Supplement of the Current Population Survey. We used biennial panel data from 2015 to 2024, focusing specifically on the proportion of disabled individuals with internet and device access. We used R to limit our data to respondents with disabilities, and filtered to only include columns for respondent state, home internet use, and device use (including mobile phone, desktop, laptop, or tablet use).

B. Statistical Models

We started by utilizing a multiple regression model to assess direct relationships between policy implementation, grant allocations, and loan amounts. This initial analysis helped identify statistically significant predictors without controlling for other factors, setting a baseline for future complex analysis (Athey & Imbens, 2016). The goal was to determine whether national policy or state-level funding correlated with increases in device loans. We chose not to include covariates, because the analysis focused on identifying basic statistically significant relationships. From these initial findings, we then implemented more complex models.

$$loans_i = \beta_0 + \beta_1 \cdot grants_i + \beta_2 \cdot policy_i + \varepsilon_i \quad (1)$$

$$grants_i = \beta_0 + \beta_1 \cdot loans_i + \beta_2 \cdot policy_i + \varepsilon_i \quad (2)$$

Equations 1 and 2 show the multiple regression models we used, where:

loans is the number of assistive technology loans,
 grants is the funding allocation, and
 policy is a binary indicator (0 = pre-policy, 1 = post-policy).

Next, we applied a two-way fixed effects model to estimate causal relationships in panel data. Fixed effect models measure only within-state, year-to-year variation, by controlling for all unobserved state conditions and national trends (Imai & Kim, 2020). We ran three fixed effects regressions to investigate different relationships pre- and post-policy: (1) effect of grants on assistive technology loans, (2) effect of assistive technology loans on internet access, and (3) effect of assistive technology loans on device access. This approach accounts for differences across states due to population size and economy, which allowed us to focus solely on various relationships between grant allocation, device loans, and internet access before and after the policy.

We standardized our grant data with a z-score to make grant funding levels comparable to loan amounts and the number of individuals with internet access. We also included an interaction term between grants and policy in two models to evaluate whether the effect of grant funding on loans changed after 2022. In the remaining models, we included an interaction term to examine how the relationship between loans and internet or digital device access changed before and after the policy. Overall, the fixed effects regressions allowed us to statistically examine the tangible effects of Assistive Technology Act-related finance changes on the disabled digital divide.

$$devices_{ist} = \beta_1 \cdot loans_{ist} + \beta_2 \cdot policy_{ist} + \beta_3 \cdot (loans_{ist} \times policy_{ist}) + \alpha_s + \delta_t + \varepsilon_{ist} \quad (3)$$

$$loans_{ist} = \beta_1 \cdot grants_{ist} + \beta_2 \cdot policy_{ist} + \beta_3 \cdot (grants_{ist} \times policy_{ist}) + \alpha_s + \delta_t + \varepsilon_{ist} \quad (4)$$

$$internet_{ist} = \beta_1 \cdot loans_{ist} + \beta_2 \cdot policy_{ist} + \beta_3 \cdot (loans_{ist} \times policy_{ist}) + \alpha_s + \delta_t + \varepsilon_{ist} \quad (5)$$

Equations 3, 4, and 5 show the fixed effects models, where:

Loans is the number of assistive technology loans in state (s), year (t),
 Internet is the number of disabled individuals with access to internet in state (s), year (t),
 Grants is the grant allocation,
 Policy is a binary indicator (0 = pre-2022, 1 = post-2022),
 Grants × policy is the interaction term, and
 α and δ are state and year fixed effects.

State programs received different amounts in grant allocations and saw different changes in assistive technology loans. Therefore, we ran a regression discontinuity model to determine the ideal grant allocation that maximized device loan amounts in the previous years. By determining the grant allocation cutoff that yields the greatest treatment effect, we aim to inform future policymaking by suggesting ideal state grant fundings to maximize device loans. Because we did not want year-to-year state variations to impact the relationship between grants and loans, we controlled for a year as a covariate. However, we acknowledge that grant allocations have different impacts on states based on other unmeasurable factors. For example, \$610,000 in funding is not enough for a large state like California, but would be more than enough for a smaller state like Rhode Island. The regression discontinuity model mainly identifies whether certain amounts of grant funding lead to statistically significant jumps in assistive technology access.

$$loans_i = \alpha + policy_{ist} \cdot (grant_i \geq 610000) + f(grant_i - 610000) + \beta \cdot year_i + \varepsilon_i \quad (6)$$

Equation 6 shows our regression discontinuity model, where:

α is the intercept,
 Policy is a binary indicator (0 = pre-2022, 1 = post-2022),
 Grant is the grant allocation,
 f represents a control for variance in loans, and
 Year is the time covariate.

C. Expected findings

Due to the focus of the 21st Century Assistive Technology Act on updating state finances, we predict that grants will be higher in the post-policy period as a result of the reauthorization. Moreover, we expect to see an increase in other variables like assistive technology loans as a result. Based on our theoretical framework, we anticipate an increase in state-financed assistive device loans due to the increased grant allocation for assistive technology. We further predict that this increase in grants and loans will also improve digital device accessibility, indirectly boosting internet usage among the disabled American population. Together, these outcomes would indicate the policy's effectiveness in improving access to assistive technology via increased funding. Our findings would pose important questions about how access to technology can improve quality of life for disabled individuals and how future policies could improve current assistive technology.

III. RESULTS

A. Descriptive statistics

California consistently received the most grant funding before and after the Act's reauthorization, while Wyoming received the least (FIG. 1). All states saw higher grant allocations after the policy, suggesting that the Act could be associated with increased funding. In contrast, changes in assistive technology loans post-policy were much less uniform across states (FIG. 2). While some states like California experienced surges in assistive technology loans, other states like Nebraska experienced substantial declines, indicating that the Act had a weaker impact on device loans.

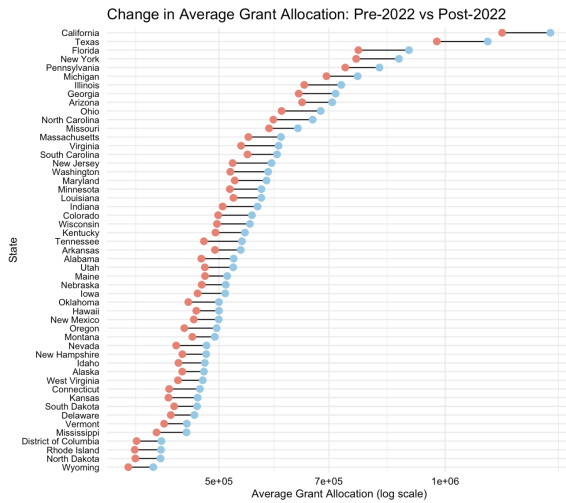


FIG 1. Dumbbell chart visualizing change in average grant allocation across states before and after the act was passed (x-axis is log-scaled).

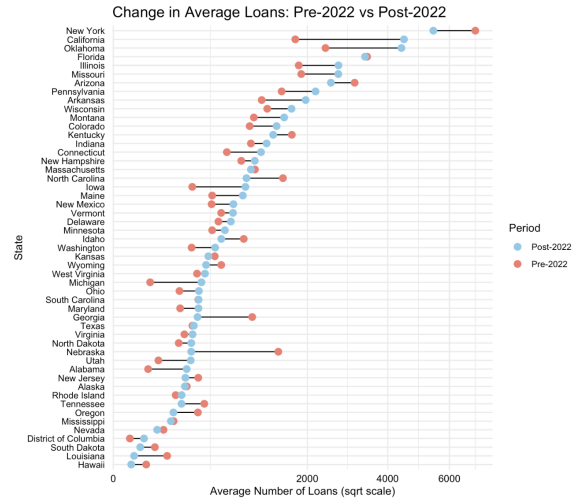


FIG 2. Dumbbell chart visualizing change in average amount of assistive technology loans prior to the reauthorization (x-axis is square-root scaled).

Descriptive statistics in Table 1 reflect similar trends in grant allocation, showing overall grant increases post-policy. Despite a nearly million-dollar gap between the maximum and minimum state grant allocations, all states saw increases in allocation post-policy.

	M	Median	SD	Min	Max
Pre-Policy (2016-2022)	536,320	489,759	153,654	363,011	1,347,585
Post-Policy (2023-2024)	590,572	538,070	180,968	410,963	1,397,328
Change (Post – Pre)	54,252	48,311	27,313	47,952	49,743

TABLE 1. Descriptive statistics (mean, median, standard deviation, minimum, maximum) values of grant allocation pre and post-policy, as well as the change between the descriptive statistics after the reauthorization of the act.

FIG. 3 and 4 demonstrate gradual increases in internet and personal computer usage among disabled individuals, with the access gap between disabled and non-disabled people decreasing over time. However, there is no sharp increase between 2021 and 2023, which might be expected following the Act's reauthorization. It is unclear whether these increases are driven by assistive technology improvements or broader global technological advancements, as both are valid reasons for those changes (Zallio & Ohashi, 2022).

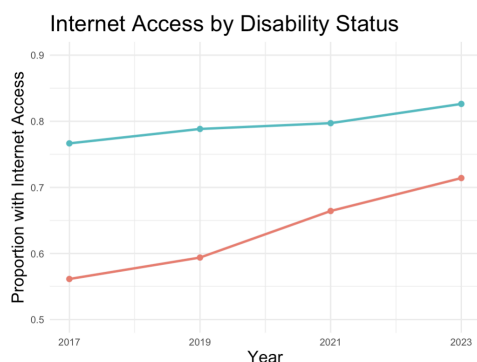


FIG. 3. This line graph shows the proportion of the U.S. population with access to the internet from 2015 to 2023.

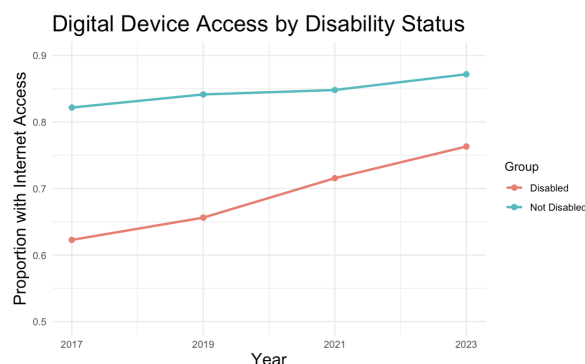


FIG. 4. This line graph shows the proportion of the U.S. population with access to a personal computer (laptop, desktop, or tablet) from 2015 to 2023.

However, these observations only reflect surface-level patterns in the data. Causal models are necessary to test and confirm the relationships between variables suggested by the data visualizations.

B. Multiple linear regression

	Grants	Loans
Intercept	-0.382*** (0.058)	956.101*** (59.009)
Policy	0.288** (0.102)	-29.743 (126.003)
Loans	0.000*** (0.000)	
Grants		504.720*** (54.432)
R²	0.185	0.171
F-statistic	51.76***	46.96***
P-value	< 2.2e-16	< 2.2e-16

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

TABLE 2. Multiple regression model summary for grants and loans with F-test.

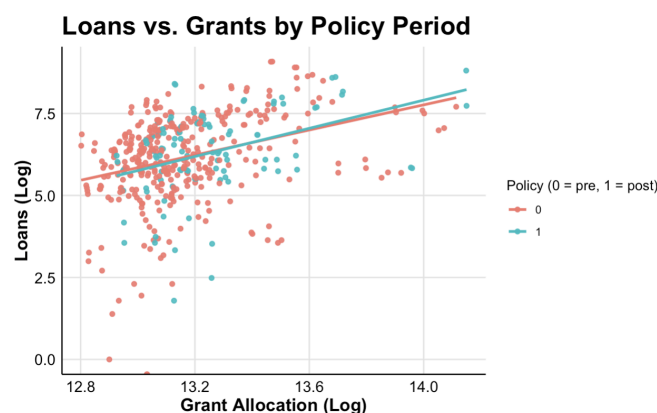


FIG. 5. Scatterplot modeling loans compared to grant allocation on a normalized log scale to account for outliers.

The first regression (Table 2, Column 1) models standardized grant allocation as a function of policy and assistive technology loans. The policy coefficient estimate indicates a statistically significant increase in grant funding post-policy ($p < 0.05$). Loan amounts are also positively associated with grant allocation. The coefficient is small, but highly significant. However, only 18.5% of variation in grants is explained by policy and loans ($R^2 = 0.185$).

The second regression (Table 2, Column 1) models loans as a function of policy and grant allocation. The policy variable is not statistically significant ($p > 0.1$), suggesting no direct policy impact on loans. However, grants have a strong effect, as the model shows that a one standard deviation increase in grants is associated with a significant increase of 505 loans ($p < 0.001$). This explains only 17.1% of the

variance in loans ($R^2 = 0.171$). The low R^2 value can partially be explained by the fact that we didn't account for confounding variables in our model.'

An F-test for both models yielded a p-value less than $2.2e-16$, well below the 0.001 threshold, confirming that both models are highly statistically significant, even without using covariates to account for confounding factors. This allows us to reject the null hypothesis that grants, policy, and loans are unrelated and proceed to more complex analyses.

C. Two-way fixed effects'

	Device loans estimate
Grants	-887.206 (557.442)
Grants × Policy	301.289* (128.488)
R2	0.877
R2 Within	0.055
RMSE	427.07

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

TABLE 3. Model summary of two-way fixed effects regression, modeling the relationship between grants and device loans during pre and post-policy periods.

The overall R^2 of 87.7% indicates that the fixed effects model explains a large portion of total variance in the U.S. However, the within R^2 is only 5.5%, showing that the model explains little variation within the state units of the fixed effects model.

In Table 3, the coefficient for grants is not statistically significant before the policy (policy = 0), but the interaction term, grants × policy, becomes statistically significant ($p = 0.023$) after the enactment of the policy. This indicates that the relationship between grants and loans changed after the reauthorization of the Assistive Technology Act, and the impact of grants on loans increased by 301 units post-policy. In summary, the policy improved the effectiveness of grant funding in increasing loan amounts.

While our linear regression did not find a strong connection between grants and loans, the more robust fixed effects model confirmed a statistically significant relationship post-policy. This adds a more complex layer of analysis to our initial hypothesis test by showing that the policy indirectly boosted assistive technology access by increasing the effect of grants on loans.

	Digital device access estimate	Internet access estimate
Loans	-5.75e-06 (4.75e-06)	-3.01e-06 (5.66e-06)
Loans × Policy	8.21e-07 (3.78e-06)	1.83e-06 (3.61e-06)
R2	0.746	0.794
R2 Within	0.003	0.001
RMSE	0.04	0.04

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

TABLE 4. Model summaries for the relationships between device loans and proportions of digital device access and internet access, during pre and post-policy periods.

We also evaluated how increases in assistive technology loans, resulting from increases in state grant allocations, impacted internet and digital device access across states.

None of the estimates are statistically significant, indicating that contrary to our expectations, no significant causal relationship exists between assistive technology access and digital access.

Furthermore, the overall and within R^2 for both the digital device and internet regression models are lower than the previous fixed effects regression with loans and grants. This tells us that assistive technology loans and the reauthorization of the Act explain less variance in digital access than grants for loans across states. The extremely small values of the coefficient estimates for the impact of loans on digital access under the policy also suggests that there is little to no relationship between loans and digital access in the first place.

D. Regression discontinuity

At a cutoff of \$610,000 in state grant allocation, we observe a statistically significant difference in the amount of assistive technology loans. Table 5 shows that states receiving funding above the cutoff issued on average 983 more device loans than states below the grant funding cutoff ($p = 0.036$). The 95% confidence interval includes 74 to 2219 loans, indicating that the model is robust. As shown in FIG. 6, the slope of the outcome variable is more shallow after the cutoff, suggesting diminishing marginal returns for assistive technology loans for most grant funding greater than \$610,000. This cutoff value was selected by plotting treatment effects across a range of possible cutoffs (see FIG. 7) and choosing the point with the maximum estimated treatment effect.

Predictor	Estimate	SE	z	p-value	95% CI	Method
Treatment above cutoff	982.92	468.92	2.1	0.036	[74.38, 2218.84]	Robust

TABLE 5. This table shows the model summary of the regression discontinuity model.

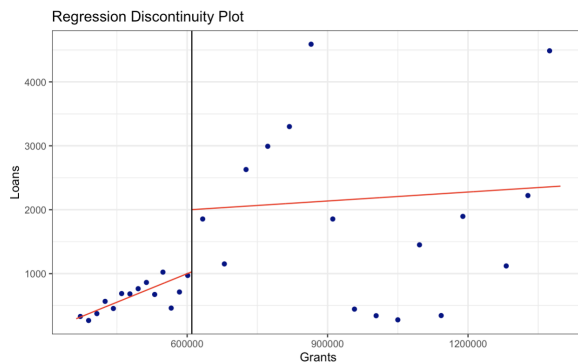


FIG. 6. This is the regression discontinuity plot for loans by different amounts of grant allocation, using triangular kernels with a cutoff at 610,000.

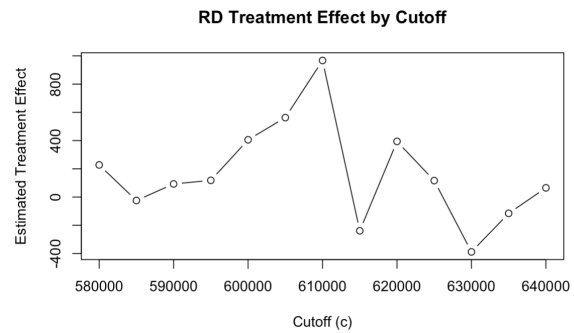


FIG. 7. This graph shows the different treatment effects by certain grant allocation cutoff values.

IV. DISCUSSION & CONCLUSION

The 21st Century Assistive Technology Act appears to have been effective in increasing grant allocations, demonstrating that the policy succeeded in its goal of improving assistive technology programs at a state level. However, our results show that the policy did not directly impact digital devices or internet access. In contrast with our hypothesis, the fixed effects models disprove a significant relationship between grants, loans, and digital access, suggesting a lack of impact of policy implementation on reducing the digital divide. From our empirical findings, it seems that the 2022 reauthorization only directly affected grant funding, indirectly affecting assistive device loans. Although current data indicates that the policy has not significantly affected loans directly, we expect a stronger trend to emerge with updated data as a more longitudinal or delayed trend. The statistically significant positive relationship between grants and loans that we identified supports this hypothesis.

We also found empirical evidence for state-level policy efficacy. Both the fixed effects and multiple regression models show that the policy improved state funding for assistive technology programs, thus increasing assistive technology loans (FIG. 8). Therefore, the policy is likely contributing to improved assistive technology access indirectly through increasing grant allocations.

However, our findings fail to support our hypothesis that the passage of the policy directly affected assistive technology loans and digital access (FIG. 8). We attribute this lack of statistically significant

relationships to a lag in national level outcomes unrepresented in currently available data. Because fewer than three years have elapsed since the policy, it is difficult to tell whether the policy actually has no significant impact or if its effects need more time to appear. However, we hypothesize that similar causal effects we observed with policy and grant funding will take place with internet and digital device access after a longer period of time. While national-level outcomes are slower to take effect, we have already observed increases in grant allocations, potentially meaning that digital technology and internet access will follow over a longer period of time (FIG. 6).

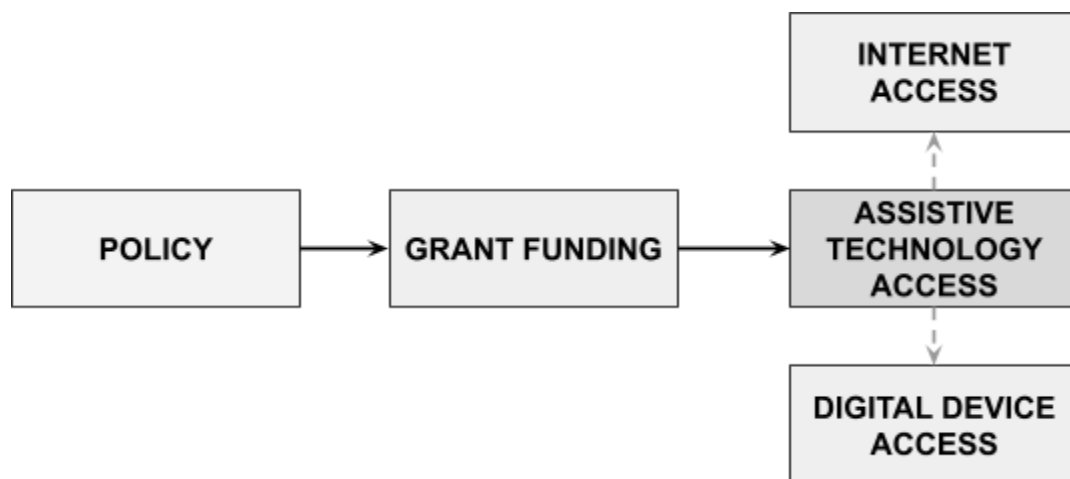


FIG. 8. Causal diagram showcasing the statistically significant relationships found through various regression analyses with three models.

Now, we return to our initial research question which sought to evaluate the efficacy of the 21st Century Assistive Technology Act. The policy made its greatest impact after the 2022 reauthorization by focusing on state-level change, primarily through expanding grant funding for individual programs. These results suggest that the government should increase efficiency of state programs before aiming to improve access on a national level.

Our results also highlight several areas of improvement. State programs should utilize allocated grant funding more effectively to improve assistive technology access, such as by including more assistive technology under Medicaid or Medicare coverage (Institution of Education Sciences, 1998). As seen in our descriptive statistics, funding has increased for all states post-policy, but the same upward trend was not observed in the data for device loans. Policymakers should aim to address disparities within states and investigate whether local conditions or administrative inefficiencies are causing these outcomes.

From our regression discontinuity analysis, we determined a cutoff of \$610,000, above which additional grant allocation yields diminishing marginal returns of assistive device loans. Based on that threshold, policymakers should consider whether allocating large amounts of funding to certain states actually leads to effective use. This could be done by focusing on the efficiency of state programs and tangible impact on device access. While outside the scope of our research, assistive technology programs vary between states, potentially meaning that some designs are more efficient or impactful than others (Myers & Dishman, 2024). More research is necessary to examine the relationship between policy, program design, and the impact on technology accessibility for individuals with disabilities.

Because the policy was reauthorized less than three years ago, clear national results are difficult to observe. Additionally, data on internet and digital access across the states is scarce, making it even more difficult to analyze policy effects. The 2022 reauthorization of the Assistive Technology Act specifically focused on increasing assistive technology access in marginalized communities such as the elderly and low income individuals, particularly elderly and low income populations. We acknowledge that the digital divide and lack of assistive technology have disproportionately impacted these populations. However, due to a lack of recent data regarding marginalized groups within the disabled community, we broadened the scope of our project to encompass the entire U.S. disabled population.

Furthermore, our research is focused on overall accessibility rather than individual impact, so we did not evaluate how the Assistive Technology Act improved physical, emotional, or psychological well-being of individuals with disabilities. This approach entails mixed-methods research, combining qualitative analysis like ours with qualitative data such as wellness surveys. Studies using methods such as factor analysis or latent class analysis could uncover underlying factors that would contribute to evaluation of the 21st Century Assistive Technology Act's efficacy.

In addition, research focusing on marginalized groups within the disabled community should be conducted to confirm that the act improves accessibility for all individuals with disabilities instead of a limited subgroup. For instance, policymakers should encourage the use of disaggregated data that focuses on smaller subgroups rather than entire populations (Ponce et al., 2023).

Most importantly, updated and longitudinal data must be collected to support future research that aims to analyze the longer-term effects of the 21st Century Assistive Technology Act. The lack of recent data post-policy means that our research was limited to evaluating immediate impacts, instead long-term outcomes. As additional data is collected relating to assistive technology, digital device, and internet accessibility, further research will be required to support or disprove a change in these longitudinal trends.

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